

# YUEZHAO LI

🔗 [liyuezhao.github.io](https://liyuezhao.github.io)

✉ [liyuezhao1996@gmail.com](mailto:liyuezhao1996@gmail.com)

## PERSONALITY

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**Nationality** Chinese  
**Date of birth** 18 April, 1996  
**Languages** Chinese (Native), English (C1), German (B1), Dutch (A1)

## RESEARCH INTEREST

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- ◇ Noncommutative geometry via spectral triples and unbounded Kasparov modules.
- ◇ Spectral localisers and computational (bivariant) K-theory of  $C^*$ -algebras.
- ◇ Topological phases of matter via groupoids and coarse geometry.

## APPOINTMENTS

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**Max-Planck-Institut für Mathematik** Mar 2026 — Aug 2026  
*Visiting postdoctoral researcher* Germany  
◇ Host: Prof. Dr. Matthias Lesch.

## EDUCATION

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**Universiteit Leiden** Oct 2021 — Feb 2026  
*PhD in mathematics* The Netherlands  
◇ Supervisor: Dr. Bram Mesland.  
◇ Thesis: *Spectral localisers and aperiodic topological phases in noncommutative geometry.*

**Georg-August-Universität Göttingen** Oct 2018 — Aug 2021  
*Master in mathematics, with minor in physics* Germany  
◇ Thesis advisor: Prof. Dr. Ralf Meyer.  
◇ Thesis: *Invariants for topological insulators coming from decompositions of coarse spaces.*

**Peking University** Sep 2013 — Jul 2018  
*Bachelor in physics, with minor in mathematics* China  
◇ Thesis advisor: Prof. Dr. Limei Xu.  
◇ Thesis: *Phase transition and critical phenomena.*

## PUBLICATIONS

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### Peer-reviewed journal articles

- [1] Y. Li and B. Mesland. *The odd spectral localiser via asymptotic morphisms and quasi-projections*. Ann. K-Theory **11**, no. 2 (2026), 213–260. [doi: 10.2140/akt.2026.11.213](https://doi.org/10.2140/akt.2026.11.213).
- [2] Y. Li. *Robustness of Topological Phases on Aperiodic Lattices*. Math. Phys. Anal. Geom. **29**, no. 2 (2026), Paper No. 16. [doi: 10.1007/s11040-026-09547-1](https://doi.org/10.1007/s11040-026-09547-1).

## Preprints

- [3] Y. Li and G. C. Thiang. *Consistent symmetry breaking and topological phases* (2026). [arXiv: 2607.28181](#).

## TALKS

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### Research talks

- ◇ AUGUST 2026: *Symmetry breaking colimit of topological phases and equivariant coarse indices*. NoSeaG 2026. Prague, Czech Republic.
- ◇ JULY 2026: *The spectral localiser via E-theory*. Frontiers in Noncommutative Geometry and its Applications. Hangzhou, China.
- ◇ MAY 2026: *Noncommutative geometry of topological phases of matter*. Higher Differential Geometry Seminar. Max-Planck-Institut für Mathematik, Germany.
- ◇ APRIL 2026: *The spectral localiser: what and why*. Oberseminar Global Analysis and Operator Algebras. Universität Bonn, Germany.
- ◇ DECEMBER 2025: *The spectral localiser via E-theory*. NCG Colloquium II. Nijmegen, Netherlands.
- ◇ NOVEMBER 2025: *The spectral localiser via E-theory*. Oberseminar Global Analysis and Operator Algebras. Universität Bonn, Germany.
- ◇ AUGUST 2025: *On topological phases of aperiodic matter*. NCG Day. Leiden, Netherlands.
- ◇ JULY 2025: *The spectral localiser via E-theory*. Oberseminar Analysis. Leibniz Universität Hannover, Germany.
- ◇ JULY 2025: *The comparison maps between groupoid and Roe  $C^*$ -algebras from aperiodic topological insulators*. Oberseminar Noncommutative Geometry. Georg-August-Universität Göttingen, Germany.
- ◇ JUNE 2025: *The spectral localiser via E-theory*.  $C^*$ -algebras, Coarse Geometry and Physics. Greifswald, Germany.
- ◇ APRIL 2025: *On robustness of topological phases* (short talk). Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Space-time. Marseille, France.
- ◇ MARCH 2025: *On robustness of topological phases of matter*. Seminar in Analysis and Applications. Delft University of Technology, Netherlands.
- ◇ JANUARY 2025: *Understanding strong and weak topological phases*. Forschungsseminar Algebra/Geometrie/Topologie. Universität Greifswald, Germany.
- ◇ SEPTEMBER 2024: *Index theory of spectral truncations and the localisation algebra*. East China Normal University, Shanghai, China.
- ◇ OCTOBER 2023: *Models of aperiodic topological insulators* (short talk). Autumn School on Large-scale Geometry. Göttingen, Germany.

### Colloquium talks

- ◇ JULY 2026: *K-theory in solid-state physics: modelling and computation*. MPI-Oberseminar, Bonn, Germany.
- ◇ SEPTEMBER 2024: *Noncommutative geometry of integer quantum Hall effect*. Sichuan University, Chengdu, China.
- ◇ MAY 2024: *A hitchhiker's guide to topological insulators in noncommutative geometry*. Leiden–Birmingham meeting, Leiden, Netherlands.
- ◇ NOVEMBER 2023: *Noncommutative geometry in index theory and physics*. Leiden–Luxembourg PhD Away Day. Leiden, Netherlands.

## CONFERENCES, WORKSHOPS AND SEMINARS

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### Participation

- ◇ AUGUST 2026: NoSeaG 2026. Prague, Czech Republic.
- ◇ JULY 2026: Summer School and International Workshop on Noncommutative Geometry. Dalian, China.
- ◇ JULY 2026: Frontiers in Noncommutative Geometry and its Applications. Hangzhou, China.
- ◇ JUNE 2026: Interfaces of Spectral Theory, Operator Algebras and Noncommutative Geometry, on the occasion of Matthias Lesch's 65 birthday. Bonn, Germany.
- ◇ DECEMBER 2025: NCG Colloquium II. Nijmegen, Netherlands.
- ◇ AUGUST 2025: NCG Day. Leiden, Netherlands.
- ◇ JUNE 2025:  $C^*$ -algebras, Coarse Geometry and Physics. Greifswald, Germany.
- ◇ APRIL 2025: Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Space-time. Marseille, France.
- ◇ OCTOBER 2024: Noncommutativity behind the Dunes. Delft, Netherlands.
- ◇ AUGUST 2024: OdenSeaG 2024. Odense, Denmark.
- ◇ MAY 2024: Leiden–Birmingham meeting. Leiden, Netherlands.
- ◇ APRIL 2024: Group Operator Algebras and Noncommutative Geometry. Marseille, France.
- ◇ NOVEMBER 2023: Leiden–Luxembourg PhD Away Day. Leiden, Netherlands.
- ◇ OCTOBER 2023: Autumn School on Large-scale Geometry. Göttingen, Germany.
- ◇ AUGUST 2023: YMC\*A. Leuven, Belgium.
- ◇ MAY 2023: Noncommutative Geometry and Operator Algebras (NSeaG 2023). Bonn, Germany.
- ◇ MARCH 2023: Noncommutative Geometry and Representation Theory. Athens, Greece.
- ◇ MARCH 2022: Noncommutativity in the North. Gothenburg, Sweden.
- ◇ FEBRUARY 2022: Winter School on Foliations, Pseudodifferential Operators and Groupoids. Göttingen, Germany.

### Organisation

- ◇ JANUARY 2023: [5th Conference of Settat on Operator Algebras and Applications](#).  
Co-organisier, with Francesca Arici, Marcel de Jeu, Rachid El Harti and Dimitris Gerontogiannis.
- ◇ I organised several [Leiden Local NCG Seminars](#) and took [notes](#) for them.
  - SPRING 2022: KK-theory ([Notes](#)).
  - FALL 2022: Groupoid  $C^*$ -algebras ([Notes](#)).
  - FALL 2023: Noncommutative geometry of foliations ([Notes](#)).

## RESEARCH VISITS

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- ◇ SEPTEMBER 2026: Universiteit Leiden, Netherlands. Hosted by Dr. Bram Mesland.
- ◇ DECEMBER 2025: Oberwolfach, Germany. Hosted by Dr. Jialong Deng.
- ◇ NOVEMBER 2025: Max Planck Institute for Mathematics, Germany. Hosted by Dr. Malte Leimbach.
- ◇ JULY 2025: Leibniz Universität Hannover, Germany. Hosted by Dr. Eske Ewert.
- ◇ JULY 2025: Georg-August-Universität Göttingen, Germany. Hosted by Prof. Dr. Ralf Meyer.
- ◇ JANUARY 2025: Universität Greifswald, Germany. Hosted by Prof. Dr. Matthias Ludewig.
- ◇ OCTOBER 2024: Peking University, China. Hosted by Dr. Guo Chuan Thiang.
- ◇ SEPTEMBER 2024: East China Normal University, China. Hosted by Prof. Dr. Hang Wang.
- ◇ SEPTEMBER 2024: Sichuan University, China. Hosted by Dr. Xingni Jiang and Dr. Yunfeng Shi.

## TEACHING EXPERIENCE

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### Universiteit Leiden

- ◇ FALL 2024: Functional Analysis, Mastermath course. Teaching assistant.
- ◇ SPRING 2024: Operator Algebras, Mastermath course. Teaching assistant.
- ◇ FALL 2023: Functional Analysis, Mastermath course. Teaching assistant.
- ◇ SPRING 2023: Differentiable Manifolds II, undergraduate course. Teaching assistant.
- ◇ FALL 2022: Functional Analysis, Mastermath course. Teaching assistant.
- ◇ SPRING 2022: Differentiable Manifolds II, undergraduate course. Teaching assistant.
- ◇ FALL 2021: Functional Analysis, Mastermath course. Teaching assistant.